

Exploring the Generative Power of Diffusion Models: From Latent Space to Multimodal Synthesis

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1. Introduction

Generative models have become one of the most influential paradigms in modern machine learning, enabling controllable synthesis of images, videos, audio, and multimodal content. Over the past few years, the field has shifted from early latent-variable models toward diffusion-based generative modeling, which now underlies state-of-the-art image generation systems such as Stable Diffusion. In this project, I systematically explore the foundations of generative modeling through a unified pipeline—from classical variational autoencoders (VAEs), to pixel-space diffusion, latent diffusion, and finally a simplified multimodal diffusion model.

I adopt the MNIST handwritten-digits dataset as a controlled environment to study the fundamental mechanisms behind representation learning and generative synthesis. Although MNIST is visually simple, it remains an ideal testbed for understanding reconstruction accuracy, latent-space geometry, denoising behavior, and conditional generation. The project begins with a VAE trained to reconstruct digits and learn a continuous latent manifold. By adjusting the KL-divergence light β , I examine the trade-off between reconstruction fidelity and latent regularity, illustrating the core tension in variational inference.

Building on the VAE, I implement both the Denoising Diffusion Probabilistic Model (DDPM) and its accelerated sampling variant, the Denoising Diffusion Implicit Model (DDIM). DDPM provides a stable training objective and high sample quality through iterative denoising, while DDIM reduces sampling steps via a non-Markovian deterministic update rule. Through empirical comparison, I highlight the efficiency–quality trade-off that characterizes diffusion-based generation.

I then integrate the trained VAE into a latent diffusion pipeline, compressing images into a compact latent space before diffusion. This mirrors the design of modern Stable Diffusion systems, where VAE compression dramatically reduces computational cost without sacrificing visual quality. Finally, I extend the pipeline to a simplified

multimodal framework by injecting CLIP embeddings into a UNet backbone, enabling text-conditioned (label-conditioned) image synthesis similar to early versions of Stable Diffusion.

To further deepen understanding of real-world diffusion workflows, I experiment with ComfyUI, a node-based system that exposes every component of the diffusion pipeline. By comparing ComfyUI’s modular graph with our tiny-mnist-sd implementation, I gain insight into how industrial-scale text-to-image systems orchestrate CLIP encoders, UNet denoisers, VAEs, and samplers in a flexible and extensible architecture.

2. Related Work

Early generative modeling was dominated by latent-variable approaches such as the Variational Autoencoder (VAE), introduced by Kingma and Welling [4]. VAEs learn a probabilistic encoder–decoder structure and optimize the evidence lower bound (ELBO) to balance reconstruction accuracy and latent-space regularization. Although VAEs provide stable training and interpretable latents, they often suffer from blurry reconstructions due to the Gaussian decoder assumption. Nevertheless, VAEs remain essential for modern diffusion pipelines, where they enable efficient latent-space compression [2].

A major breakthrough came with diffusion models. Ho et al. [3] formalized DDPM as a stochastic reverse process that iteratively denoises Gaussian-corrupted data. DDPM models achieve competitive sample quality but require hundreds of sampling steps. To address this limitation, Song proposed DDIM [7], introducing a non-Markovian deterministic sampler that dramatically speeds up generation while maintaining visual fidelity. Diffusion processes are typically implemented using a UNet-based denoiser [6], which serves as the core architecture in both research prototypes and production-grade systems.

The next major milestone in generative modeling is latent diffusion. Instead of operating directly in pixel space,

Esser et al. and later RFT models [2] compress images via a VAE into a lower-dimensional latent space where diffusion is performed. This approach enables high-resolution synthesis at a fraction of the computational cost and provides the architectural foundation for Stable Diffusion. Latent diffusion pipelines also integrate cross-modal conditions—most notably CLIP [5], which aligns text and image embeddings in a shared space. By injecting CLIP embeddings into the UNet, diffusion models gain controllable text-to-image synthesis capabilities.

In practice, modern diffusion systems are orchestrated via modular toolkits such as ComfyUI [1], which expose internal components (CLIP, UNet, VAE, samplers) as configurable nodes. ComfyUI reflects the structure of industrial-scale pipelines and serves as an educational bridge between conceptual algorithms and real-world implementations.

3. Experiments

This section presents two phases of experiments: the **Basic Task** and the **Advanced Task**.

3.1. Basic Task

VAE: Effect of KL Weight. We first study the influence of the KL-divergence weight β on VAE reconstruction and latent sampling. Four configurations are evaluated: $\beta = 0.1$, 0.5, 1.0, and 5.0, each trained for 40 epochs. Figure 1 and Figure 2 summarize representative reconstruction and sampling behaviors.

When β is small (e.g., $\beta = 0.1$), the KL term contributes very weak regularization. The latent space becomes irregular and highly overfitted to the training set. This leads to extremely sharp reconstructions but unstable sampling, as random latent vectors do not align with the distribution learned by the encoder. A moderate value such as $\beta = 0.5$ produces the best trade-off. Reconstructions remain sharp, while latent samples exhibit good diversity and stable digit shapes. This indicates that the latent manifold is both expressive and sufficiently regularized. When β increases to 1.0, reconstruction quality begins to deteriorate. The model places more emphasis on matching the latent distribution to a standard normal prior, resulting in slightly blurred edges and reduced detail. Sampling becomes more stable but visually less diverse.

At an extreme such as $\beta = 5.0$, the model becomes over-regularized. The encoder collapses towards the origin, causing heavy information loss. Reconstructions appear blurry and lack fine structure, while sampled images deteriorate significantly. The results confirm classical VAE behavior: small β values produce crisp reconstructions but poor generative quality, while excessively large β values lead to overly smoothed reconstructions. In practice, $\beta \in [0.3, 0.7]$ yielded the most stable balance for MNIST in our experiments.

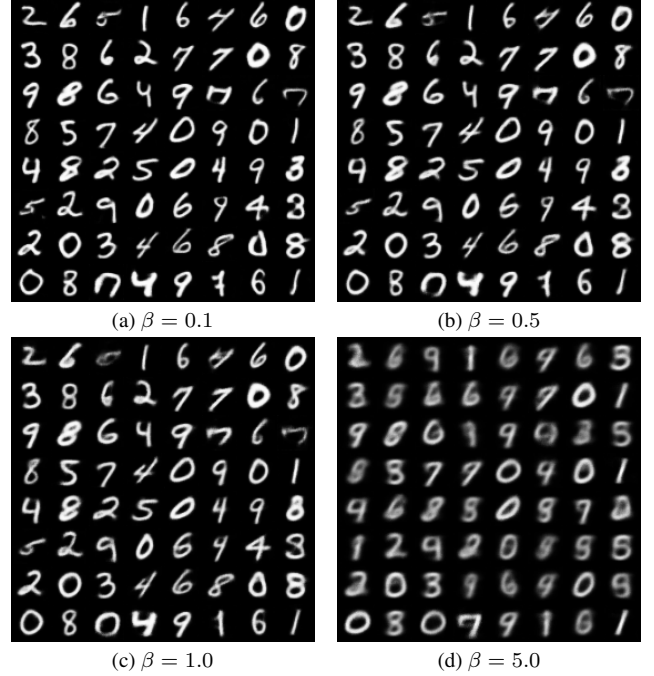


Figure 1. VAE reconstruction comparison under different β values at epoch 40. Smaller β yields sharper reconstructions, while larger β leads to oversmoothing.

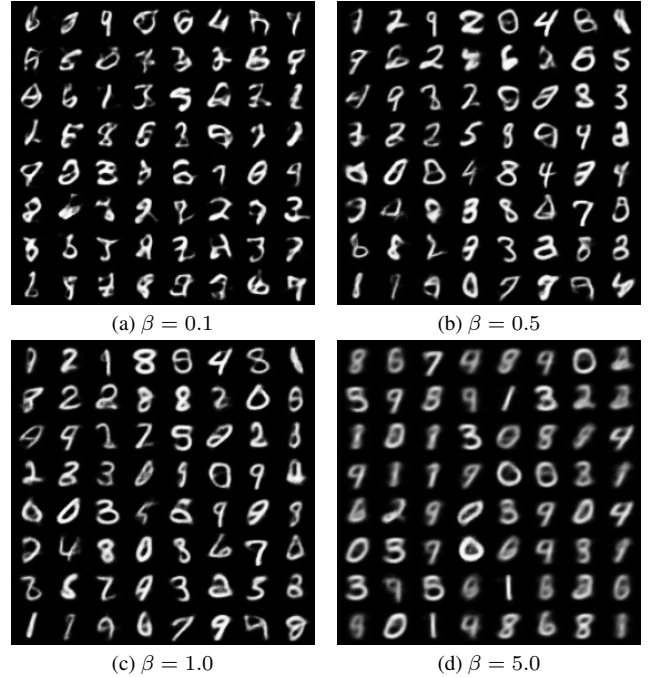


Figure 2. Random samples from VAEs trained with different β values. Moderate β (e.g., $\beta = 0.5$) achieves a better balance between sample diversity and visual fidelity.

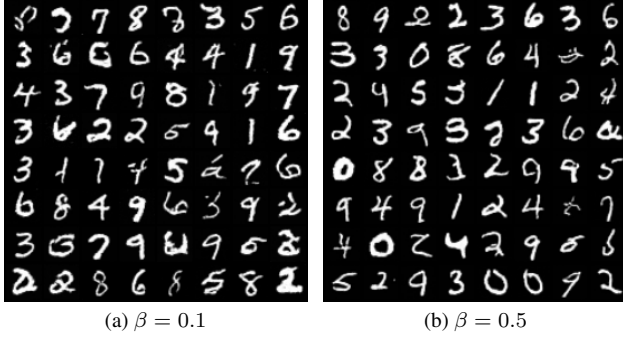


Figure 3. Qualitative comparison between DDPM and DDIM sampling on MNIST. DDIM preserves global digit structure while requiring significantly fewer sampling steps.

DDPM vs. DDIM. We next compare DDPM and DDIM sampling under the same training conditions. The diffusion model is trained for 20 epochs with 1000 training timesteps, and DDIM sampling is tested using 200 deterministic steps with $\eta = 0$.

Training loss steadily decreases from 0.0768 to 0.0167, indicating stable denoising learning. Qualitatively, DDPM samples exhibit clear digit shapes but require many sampling steps. DDIM significantly accelerates sampling while maintaining comparable visual quality. The deterministic update rule of DDIM preserves global digit structure, although slight artifacts occasionally appear due to fewer integration steps.

These observations match theoretical expectations: DDPM provides high-quality sampling via stochastic refinement, whereas DDIM offers a computationally efficient alternative with minimal degradation when applied to simple datasets like MNIST.

Latent Diffusion. We further train latent diffusion models using VAEs with $\beta = 0.5$ and $\beta = 1.0$. Latent-space training exhibits significantly higher reconstruction loss compared to pixel-space diffusion, as latent features are more compressed and structured. Across 40 epochs, both models converge to a stable loss around 0.28.

Although the losses are numerically higher, the visual quality of latent-based samples is competitive and often smoother compared to pixel-space diffusion. Among the two VAEs, $\beta = 0.5$ consistently yields better latent diffusion results: the latent space is compact yet expressive enough to allow visually coherent samples. In contrast, $\beta = 1.0$ leads to under-expressive latent codes and slightly degraded generation quality.

These experiments confirm the practicality of latent diffusion even in constrained datasets, and they align with observations from modern image synthesis pipelines such as Stable Diffusion.

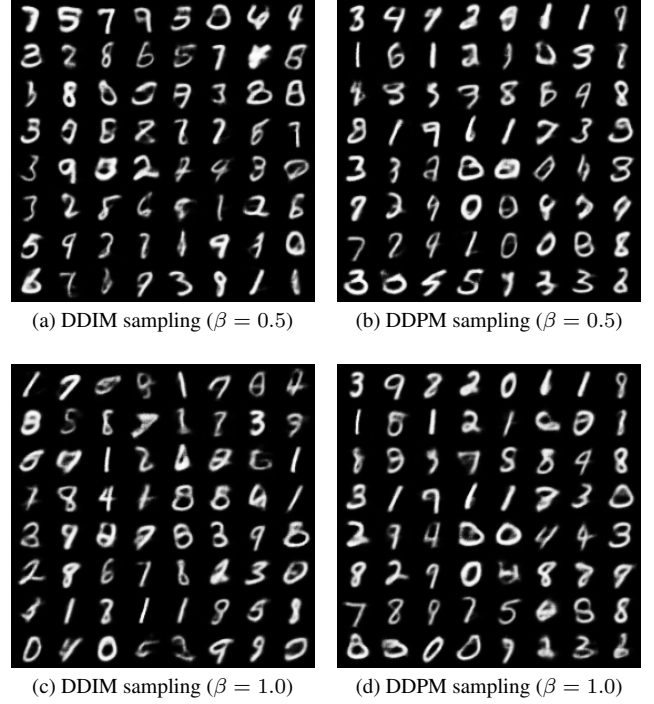


Figure 4. Comparison of latent diffusion sampling under two VAE configurations ($\beta = 0.5$ and $\beta = 1.0$), using both DDIM and DDPM samplers. Models trained with $\beta = 0.5$ produce more coherent and visually stable digits, while $\beta = 1.0$ exhibits degraded structure due to an under-expressive latent space.

3.2. Advanced Task

3.2.1. Tiny MNIST Stable Diffusion

Training a CLIP Encoder on MNIST. To enable conditional generation, we first train a lightweight CLIP model that aligns MNIST digit images with their corresponding label embeddings. As shown in Figure 5, the training process converges rapidly within the first ten epochs: the contrastive loss drops sharply while validation accuracy rises above 97%. Extending training to fifty epochs further improves performance, reaching a test accuracy of 98.57%. These results indicate that the contrastive objective effectively captures discriminative visual patterns on MNIST.

However, the model’s zero-shot generalization ability remains extremely weak. As summarized in Table 1, the zero-shot accuracy is only 15.62%, despite near-perfect supervised accuracy. This reveals that the learned embedding space primarily reflects *pixel-level correlations* rather than meaningful semantic alignment between the text and image modalities. Consequently, text embeddings (digit labels) do not form well-separated clusters, limiting the model’s ability to generalize to unseen text prompts. This deficiency in cross-modal alignment later becomes a bottleneck for the conditional diffusion model.

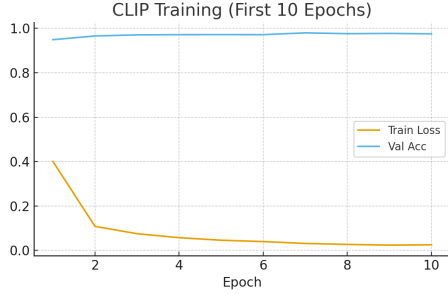


Figure 5. Training dynamics of the CLIP encoder on MNIST. The model converges rapidly within the first 10 epochs, achieving high validation accuracy while steadily reducing contrastive loss.

Table 1. CLIP supervised vs. zero-shot performance on MNIST.

Metric	Accuracy
Validation Accuracy (supervised)	98.40%
Test Accuracy (supervised)	98.57%
Zero-shot Accuracy	15.62%

Conditional Tiny MNIST Stable Diffusion. Using the trained CLIP encoder, we then train a small Stable-Diffusion-style UNet conditioned on CLIP embeddings. The training loss curve in Figure 6 shows steady convergence toward approximately 0.276 over 40 epochs. Initial generations, however, were blurry and misaligned with their conditioning labels. After correcting the DDIM sampling implementation—specifically the computation of x_0 and the deterministic update rule—sampling quality improved substantially.

Figure 7 illustrates this improvement: the corrected sampler produces noticeably clearer and more coherent digits. Nevertheless, even with the corrected sampling procedure, the model still struggles to generate consistent columns of digits (0–9). This limitation is consistent with the earlier CLIP findings: because the CLIP encoder provides weak semantic separation between digit labels, the conditional UNet receives insufficiently informative guidance, resulting in unreliable label-conditioned generation. Despite these challenges, both the 40-epoch and extended 200-epoch models occasionally produce visually plausible samples, demonstrating partial controllability. These findings highlight the critical role of a semantically aligned text–image embedding space in enabling robust conditional diffusion.

Sampling Correction and Improved Generations. During early experiments, the conditional diffusion model consistently produced blurry or semantically inconsistent samples, even after extended training. This led us to suspect

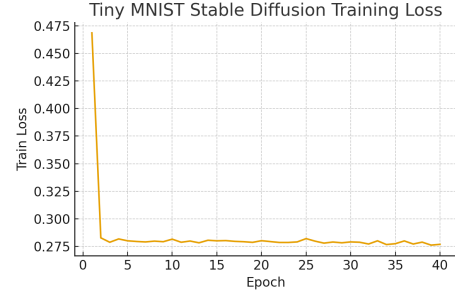


Figure 6. Training loss of the Tiny MNIST Stable Diffusion model over 40 epochs. The loss stabilizes around 0.276, consistent with convergence of a lightweight conditional UNet.

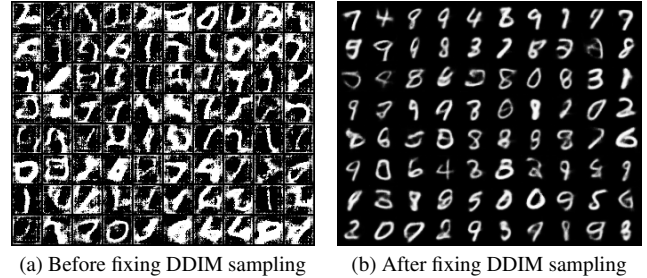


Figure 7. Comparison of Tiny MNIST Stable Diffusion outputs before and after correcting the DDIM implementation. The corrected model produces significantly clearer and more coherent digit shapes.

an implementation issue in the sampling procedure. Upon further inspection, two problems were identified: (1) the conditioning vector was not concatenated correctly with the time embedding and latent input inside the UNet, and (2) the DDIM update rule in the sampler did not correctly follow the deterministic transition equations, particularly the computation of x_0 and the noise-free update when $\eta = 0$.

After correcting these issues—as shown in the code block above, where we explicitly concatenate $[x, \text{time_emb}, \text{cond}]$ in the UNet input channel and properly compute the DDIM coefficients—sampling quality improved substantially. To illustrate this, we present several representative samples from two training stages.

From the 40-epoch model, Figures 8(a)–(b) show two of the clearest outputs. Although not perfectly aligned with their conditioning labels, these samples reflect noticeably sharper digit shapes and cleaner stroke boundaries compared to the pre-fix outputs.

Extending training to 200 epochs yields further improvements. Figures 8(c)–(e) display three samples that demonstrate more distinct digit structures and higher visual stability. Even though perfect conditional alignment is still not achieved—largely due to the weak semantic separation in the CLIP embeddings—the results show clear progression

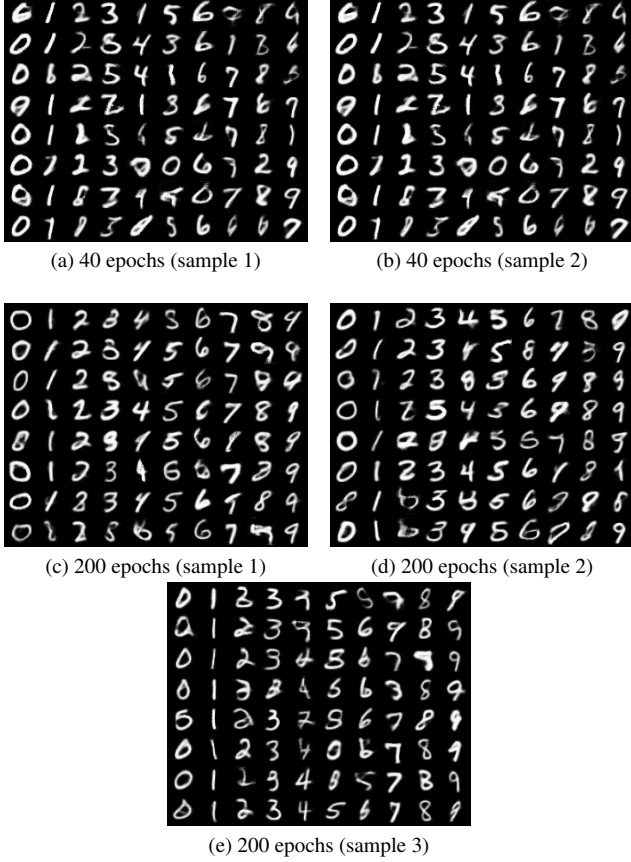


Figure 8. Representative samples after correcting the conditional DDIM sampling implementation. Top row: two selected outputs from the 40-epoch Tiny MNIST SD model. Bottom row: three outputs from the 200-epoch model, showing progressively clearer and more stable digit structures.

in visual quality, confirming the positive impact of the corrected sampling method.

3.2.2. ComfyUI Exploration

To further understand modern diffusion pipelines beyond our tiny MNIST Stable Diffusion implementation, we explore ComfyUI, a node-based generative workflow system. This platform exposes the modular components of large-scale diffusion models and allows us to experiment with checkpoint loading, prompt engineering, LoRA integration, and model customization. Through these experiments, we aim to analyze how well state-of-the-art text-to-image models handle highly constrained numerical generation tasks compared to our MNIST-focused models.

Environment Setup. We install ComfyUI in a clean environment and launch the system in server mode. The interface is accessible via a local browser, presenting a visual graph editor. ComfyUI organizes model files under the

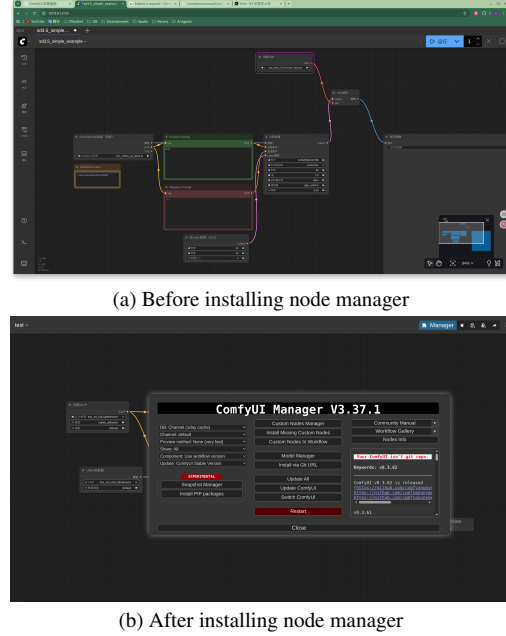


Figure 9. User interface before and after enabling custom node support in ComfyUI.

models/ directory, including checkpoints, VAE decoders, CLIP text encoders, LoRA modules, embeddings, ControlNet files, and upscalers. To test compatibility, we import our own project checkpoints (VAE, CLIP, DDPM/DDIM, and tiny SD models) into the corresponding folders, though only standard-format components can be loaded without additional customization.

Custom Nodes for Loading Project VAEs. ComfyUI’s default VAE node supports only Stable Diffusion-compatible weights. Our MNIST VAE, trained with a different architecture, cannot be loaded with built-in nodes. To address this, we install a node manager that enables custom node extensions. After installation, the interface updates (Figures 9a–9b), confirming that the system now supports customized model logic.

Using the provided `example_node.py.example` template, we can construct a dedicated loader for our MNIST VAE format. This mirrors real-world scenarios where diffusion pipelines must integrate heterogeneous models, demonstrating ComfyUI’s flexibility and its alignment with modular generative system design.

Default Workflow Experiments. After successfully importing standard Stable Diffusion checkpoints, we first execute ComfyUI’s default text-to-image workflow. Figure 10a shows an example of the initial output.

To test whether a general-purpose diffusion model can generate structured numerical grids, we apply the following

prompt: generate 8×10 matrix, from zero to nine numbers, a black and white image, and with a negative prompt discouraging unwanted content: people and object, blur.

The resulting output (Figure 10b) shows weak adherence to the requested structure: digits appear but the grid is incoherent and lacks clear alignment.

To improve control, we design a stronger prompt emphasizing constraints on shape, contrast, repetition, and visual clarity: Black-and-white bitmap, 8×10 matrix of Arabic numerals 0–9, each column contains 8 different digits without repetition, high-contrast white background, crisp black digits, pure monochrome, 1-bit depth, no grid lines, no shading, no color, and with a stronger negative prompt suppressing all non-numerical elements: No color, no grayscale, no grid, no shadow, no anti-aliasing, no blur, no distortion, no decorative elements, no repetition within any column.

These stronger prompts yield visually improved results (Figures 10c–10d), but still fail to satisfy structural constraints. While digits occasionally appear correctly, they are not arranged in consistent grid patterns or column-wise label control as requested.

This limitation arises because the default workflow uses a Stable Diffusion v1.5 checkpoint (v1-5-pruned-emaonly-fp16.safetensors), a model optimized for natural image synthesis rather than precise pixel-level symbolic generation. Such models emphasize diversity, color richness, and semantic composition, making them poorly suited for generating black-and-white numerical layouts.

LoRA-Enhanced Workflows. We test LoRA-augmented workflows to evaluate whether style-conditioning modules can improve numerical generation. Figure 11 shows a multi-LoRA setup.

However, most LoRAs available in the ecosystem are trained for stylistic or character-centric domains (e.g., anime, portrait styles, scenery). Even when combining multiple LoRAs and strengthening negative prompts, the model continues to prioritize human-like shapes or stylistic elements. Several partially usable outputs are shown in Figures 12a–12d, but overall, the LoRA modules bias the model toward generating figures rather than clean digit matrices.

This behavior highlights the essential limitation of style LoRAs: they modulate the model’s output distribution toward their training domain, making them unsuitable for tasks requiring strict pixel-level symbolic control, such as MNIST-like digit layout generation.

4. Conclusion

This work systematically explores generative modeling on MNIST across three levels of complexity: basic latent-variable models, diffusion-based sampling strategies, and

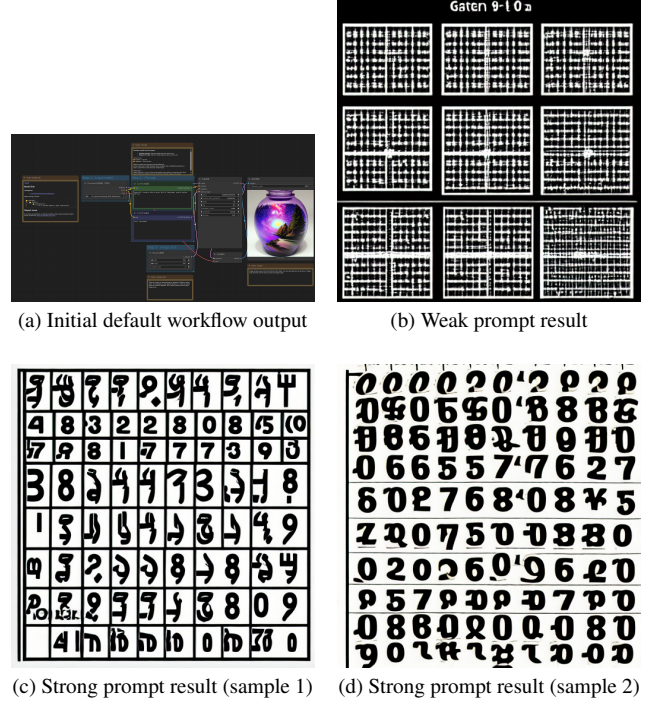


Figure 10. Outputs generated by the default Stable Diffusion workflow in ComfyUI. While digits appear under stronger prompt constraints, the model does not reliably produce structured MNIST-like numerical grids.

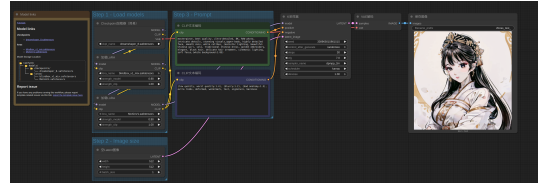


Figure 11. Multi-LoRA workflow configuration in ComfyUI.

conditional generation pipelines inspired by Stable Diffusion. The experiments reveal several key findings regarding model behavior, architectural constraints, and the role of cross-modal alignment.

In the **Basic Task**, we observe that the choice of VAE regularization weight β is crucial for balancing reconstruction fidelity and latent expressiveness. Very small β values ($(\beta=0.1)$) produce sharp reconstructions but fail to support stable generative sampling due to an irregular latent space. Excessively large values ($(\beta=5.0)$) over-regularize the encoder, degrading both reconstructions and sampled digits. Moderate values near $(\beta=0.5)$ offer the best trade-off, yielding clean reconstructions and diverse, coherent samples. Similarly, a comparison of DDPM and DDIM shows that deterministic DDIM sampling preserves overall digit structure with far fewer steps, confirming theoretical

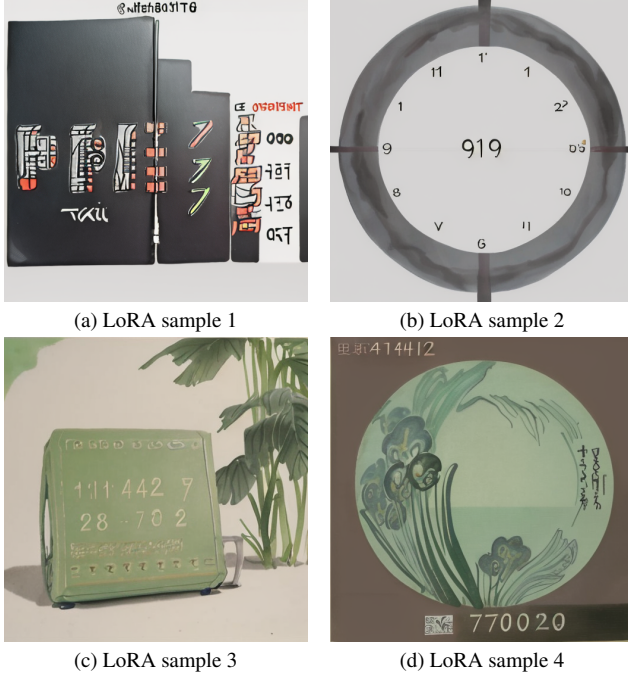


Figure 12. Selected outputs from LoRA-enhanced workflows. Despite strong negative prompting, LoRA modules heavily bias generation toward human-like forms, demonstrating unsuitability for structured numerical tasks.

expectations of improved efficiency with minimal quality degradation. Latent diffusion further demonstrates that diffusion models can operate effectively in compressed latent spaces, provided that the underlying VAE produces sufficiently expressive representations; again, $\beta = 0.5$ outperforms $\beta = 1.0$ in this respect.

In the **Advanced Task**, we build a conditional Tiny MNIST Stable Diffusion pipeline using a lightweight CLIP encoder. While supervised CLIP accuracy exceeds 98%, zero-shot accuracy remains extremely low (15.62%), revealing a fundamental limitation: the learned image-text embedding space captures pixel-level correlations rather than semantic alignment. This misalignment directly impacts the conditioning mechanism of the diffusion model. Even after correcting the DDIM implementation and improving sampling quality, the conditional UNet fails to maintain column-wise digit consistency or reliably follow label prompts. Extended training (up to 200 epochs) improves visual clarity but does not overcome the underlying representation bottleneck. These results highlight that the success of conditional diffusion hinges not merely on the UNet or sampler, but critically on the semantic structure of the embedding space used for conditioning.

Our exploration of **ComfyUI** further reinforces these insights. Despite using powerful Stable Diffusion v1.5 checkpoints and extensive prompt engineering, the default

workflow cannot produce structured MNIST-style numeric grids. The model’s inductive bias—optimized for natural images, color richness, and semantic diversity—prevents it from generating precise black-and-white symbolic patterns. LoRA-augmented workflows amplify this mismatch, as community LoRAs are overwhelmingly trained on stylized domains (e.g., anime, portraits), resulting in persistent human-like artifacts even under strong negative prompts. These findings show that state-of-the-art text-to-image diffusion models are poorly suited for strict symbolic generation tasks without domain-specific training or architectural modification.

Overall, the experiments suggest several broader conclusions. First, expressive yet well-regularized latent spaces are essential for stable generative modeling, both in VAEs and diffusion systems. Second, high-quality conditional diffusion requires a semantically meaningful embedding space; strong supervised accuracy alone is insufficient. Third, large-scale text-to-image models exhibit inherent biases that limit their applicability to symbolic or structured-generation domains. These observations provide practical guidance for constructing diffusion pipelines aimed at controlled, label-aligned, or symbolic image synthesis: improving the embedding space, rather than solely modifying the sampler or UNet, is likely to yield the most significant gains.

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